

WASP-43b

R-0630

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_180105 · created 2026-07-25T15:33:10+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458123.8517 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.016
Transit depth (Rp/Rs)^2	2.92 +/- 0.56 [%]
Orbital Inclination (inc)	80.88 +/- 0.98
Airmass coefficient 1 (a1)	1.185 +/- 0.016
Airmass coefficient 2 (a2)	-0.12 +/- 0.0048
Scatter in the residuals of the lightcurve fit is	1.36 %
Best Comparison Star	None
Optimal Aperture	3.73
Optimal Annulus	15.94
Transit Duration (day)	0.0473 +/- 0.004

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.171 ± 0.016 vs 0.1594 (deviation 0.63σ)
Duration fit vs published	0.0473 d vs 0.0512 d (deviation 0.86σ)
Mid-time O–C	1.0 min
Depth SNR	9.3
Residual scatter	1.36 %

Light curve

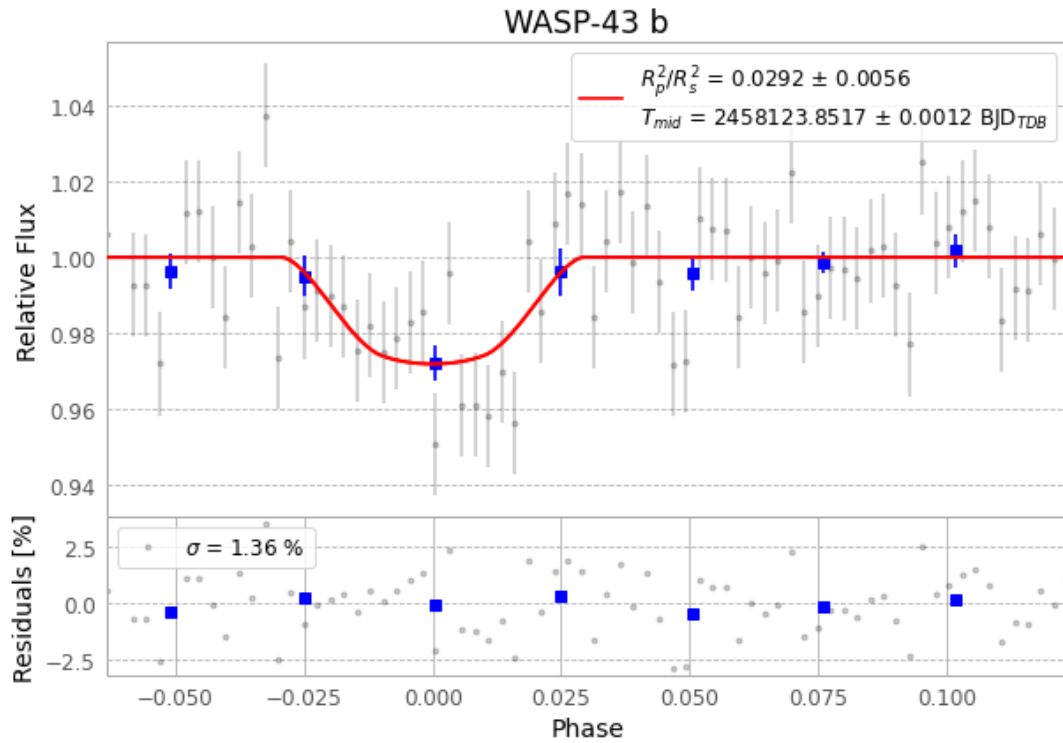


Figure 1 — FinalLightCurve_WASP-43 b_2018-01-05.png (SHA-256 57e05a7f3057459b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [361, 233], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f0a3fddda73982a9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

74 FITS frames (49 MB), WASP-43180105070612.FITS → WASP-43180105104512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-180105.log (9c35b897ec8b...), FinalLightCurve_WASP-43 b_2018-01-05.png (57e05a7f3057...), FinalLightCurve_WASP-43 b_2018-01-05.csv (896385b1ae5a...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)